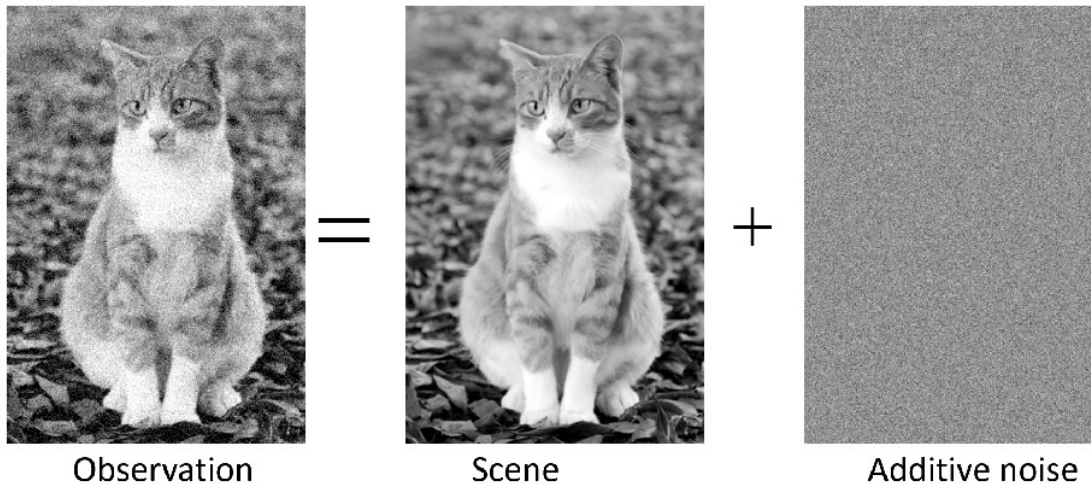


A Glance of Image Denoising

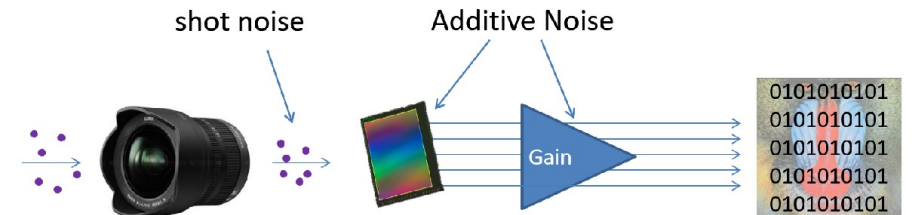
Guo Qilong

Image Denoising

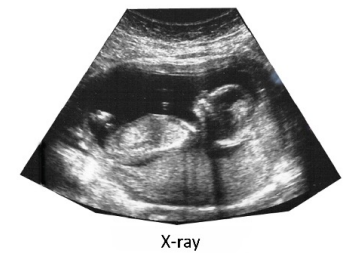
- Primary problem of inverse problem: $y = Ax + \epsilon$, where $A = I$, x is the true value, y is the noisy observation, ϵ is the noise.
- Source: Sensor noise, photon counting statistics, low-light conditions, transmission and compression errors.



- Sources of noise



- Examples



Statistical Model of Noise

- Noise is usually modeled as a random vector that follows a probability distribution.
- Poisson noise: Quantum noise caused by the randomness of photo arrivals. The noise vector is i.i.d. whose entries follow a discrete Poisson density function:

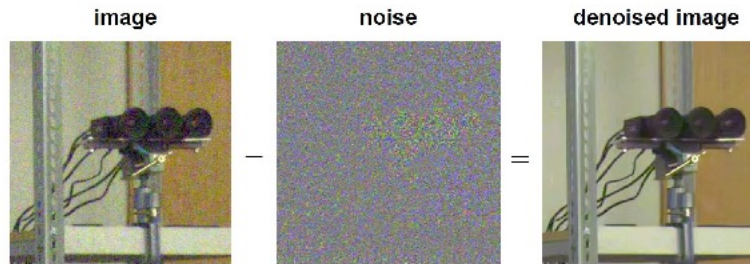
$$\epsilon(x) \sim Poi(\rho, T)$$

- Gaussian noise: Thermal noise caused by electronic imperfection. The noise vector whose entries follow a normal density function:

$$\epsilon \sim N(0, \sigma^2)$$

Evaluation

- Assume $\tilde{y}$ is denoised result from noisy image y .



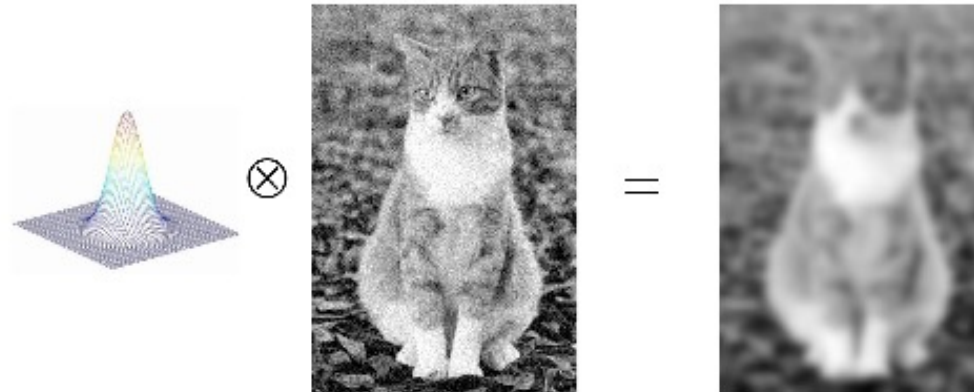
- Quality measurement of the denoised result $\tilde{y}$:
- Mean Square Error (MSE): $MSE = Mean(||x - \tilde{y}||_2^2)$.
- Peak Signal-to-Noise Ratio: $PSNR = 10 * \log_{10} \left(\frac{255^2}{MSE} \right)$.

Traditional Denoising Method

- Linear methods:
 - Smoothing by a low-pass filter (like Gaussian filter)
 - Wiener filtering (optimal linear method)
- Nonlinear filtering methods:
 - Median filter (take the median within local values).
 - Bilateral filter (filter consists of two factors: spatial proximity (how close), and pixel value similarity (how similar intensities)).
- Transform thresholding method:
 - Wavelet thresholding method (Denoises by suppressing small coefficients in an orthogonal and complete multiscale wavelet basis, exploiting **signal sparsity**).
- Non-local method:
 - Non-local mean (weighted average of similar patches)
 - BM3D (Block-Matching and 3D Filtering) (groups similar patches, stacks them into 3D volumes, and performs thresholding in the transform domain.)

Removing Noise by Smoothing

- Motivation: assuming noises to be i.i.d. with zero mean, therefore the average of nearby pixels can lower the variance.
- E.g.: convolution by gaussian filter:



Wiener Filtering: Optimal Linear Denoising

- Smoothing filter leads to blurry results.
- Convolution theorem:

$$\tilde{y} = h \otimes y \Leftrightarrow \hat{\tilde{y}}[\omega] = \hat{h}[\omega] \cdot \hat{y}[\omega] = \hat{h}[\omega] \cdot (\hat{x}[\omega] + \hat{\epsilon}[\omega])$$

- Wiener filtering: Find a filter h such that for each ω ,

$$\hat{\tilde{y}}[\omega] = \hat{h}[\omega] \cdot \hat{y}[\omega]$$

is as close as to the true $\hat{x}[\omega]$.

- Derivation: optimize expectation of mean square error (Wiener estimator, i.e., linear operator that minimize the bayesian risk):

$$\min_{\hat{h}} E\{||\hat{\tilde{y}} - \hat{x}||_2^2\} = \min_{\hat{h}} E\{||(1 - \hat{h})\hat{x} - \hat{h}\hat{\epsilon}||_2^2\}$$

Derivation of Wiener Filter

- Assume (mostly consider Additive White Gaussian Noise(AWGN)):
 - **Signal are i.i.d, and noise are i.i.d**
 - **Signal and noise are independent**
 - **Noise with zero means**

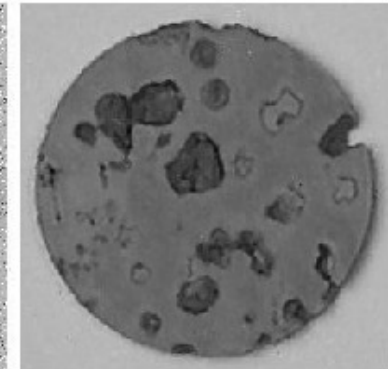
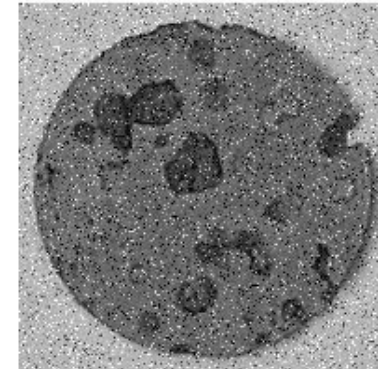
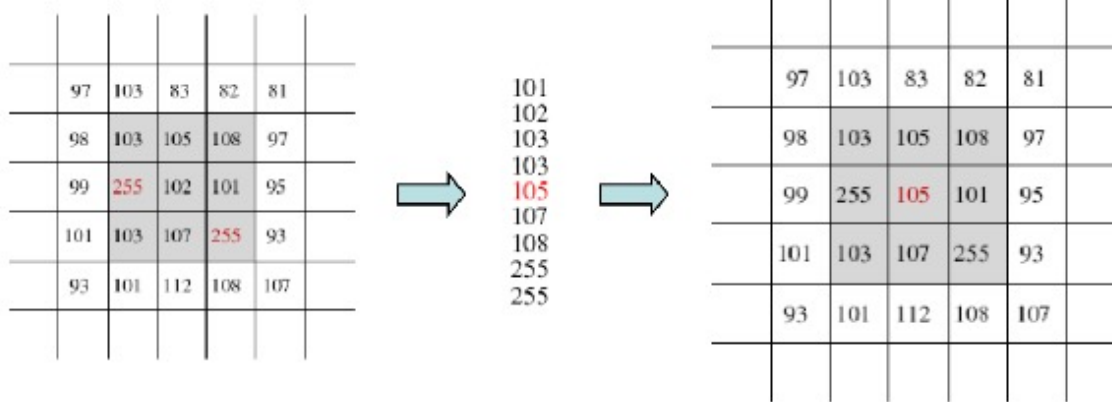
- Therefore, Wiener filter is given after calculation:

$$\hat{h}[\omega] = \frac{E\{|\hat{x}[\omega]|^2\}}{E\{|\hat{x}[\omega]|^2\} + E\{|\hat{\epsilon}[\omega]|^2\}} = 1 - \frac{1}{1 + \frac{E\{|\hat{x}[\omega]|^2\}}{E\{|\hat{\epsilon}[\omega]|^2\}}}$$

- The Wiener filter is more for theoretical understanding, it is not practical in general to be exactly implemented for denoising as it requires to know the truth.

Median Filtering

- Salt-and-pepper noise is a type of image noise where some pixels are randomly set to maximum (white, “salt”) or minimum (black, “pepper”) intensity values.



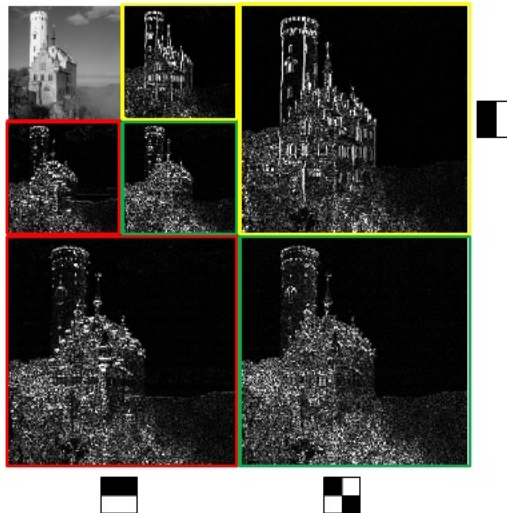
Wavelet Thresholding

- Implement threshold to get sparse representation of signal
 $y \Rightarrow Wy \Rightarrow \Gamma(Wy) \Rightarrow \tilde{y} = W^{-1}\Gamma(Wy)$

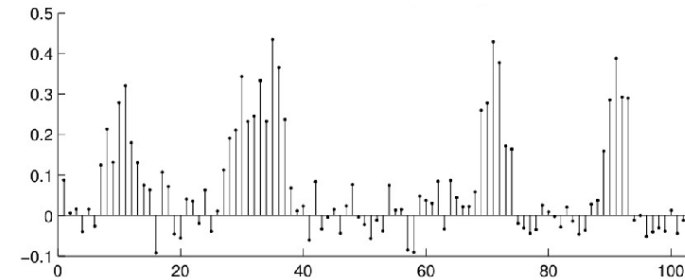
Original image



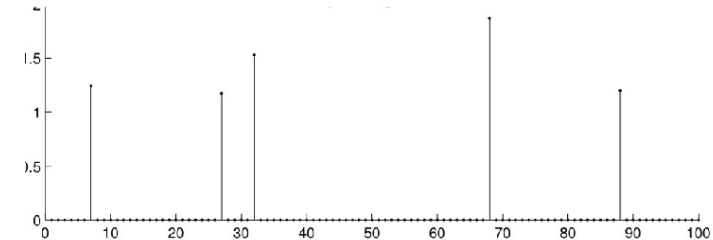
2 level DWT



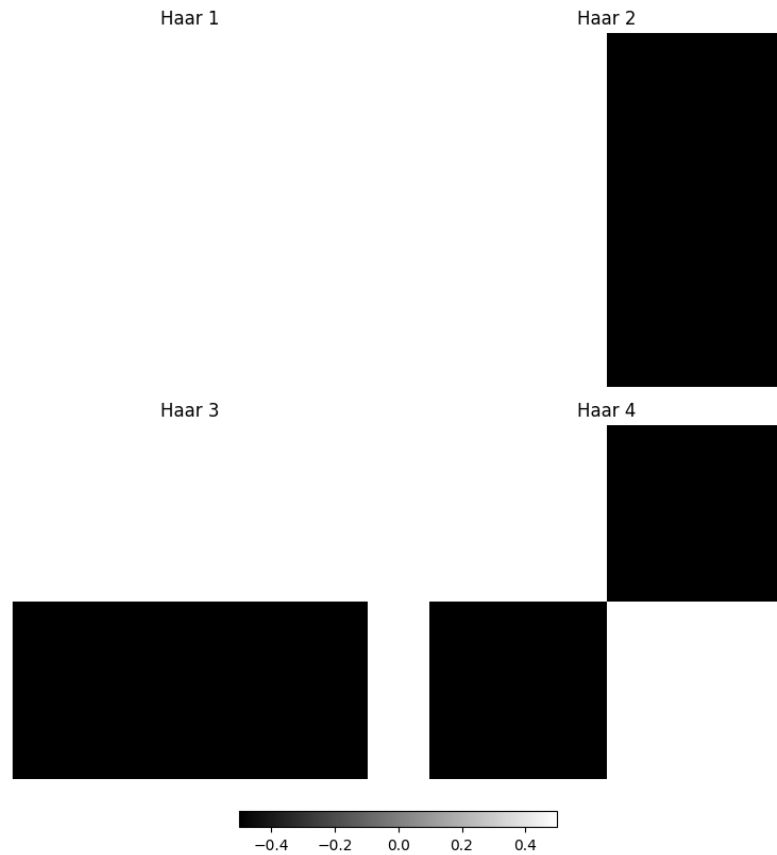
Signal f (non-sparse)



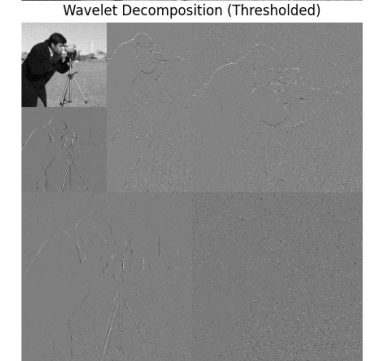
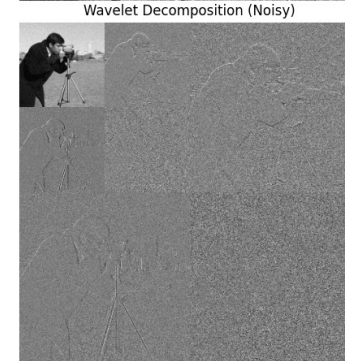
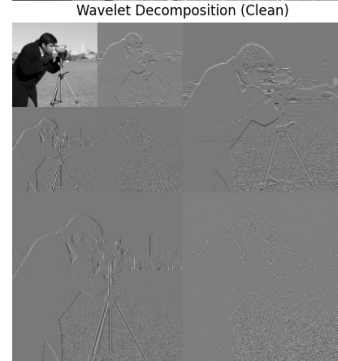
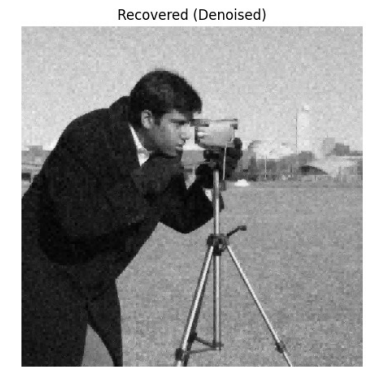
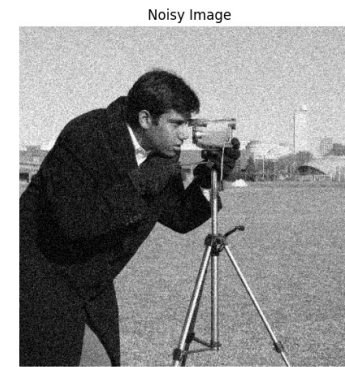
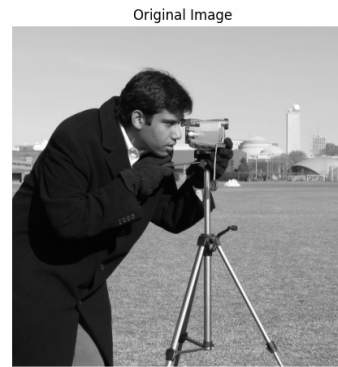
Transform coefficients Wf (sparse)



Example by Haar Filter Banks



Wavelet Denoising Example



Supervised Image Denoising

- Supervised methods train neural networks (NNs) $f(\cdot)$ on paired datasets $\{(\mathbf{x}, \mathbf{y})\}$, which often are synthesized w.r.t. Additive White Gaussian Noise (AWGN).

$$L_{supervised} = E\{||f(\mathbf{y}) - \mathbf{x}||\}$$

- Denoising NNs trained on such synthetic noise often fail to perform well on real-world data due to significant differences of AWGN and real-world noise.
- The creation of real-paired datasets is labor-intensive and often fails to cover the full spectrum of real-world noise. Also, ground-truth (GT) images are difficult to obtain in fields like medical or scientific imaging.
- Unpaired dataset learning also require clean datasets.

Self-supervised Image Denoising with Only Noisy Images

- Remove the requirement of clean images.
- Carefully design the loss function to make it equivalent with supervised loss.
- Assume only with noise: zero-mean, iid.
 - Noise2Noise [1]: paired noisy images $\{(\mathbf{y}_1, \mathbf{y}_2)\}$, $\mathbf{y}_1 = \mathbf{x} + \mathbf{n}_1, \mathbf{y}_2 = \mathbf{x} + \mathbf{n}_2$
$$E\{\|f(\mathbf{y}_1) - \mathbf{y}_2\|_2^2\};$$
 - Data augmentation: using simulated paired noisy images to remove the need of multiple captures of noisy images in the same scene. Such as R2R [2] ZS-N2N[3]
 - Blind Spot Scheme: using neighboring noisy pixels $\mathbf{y}_{\setminus i}$ to predict the hidden pixel y_i , i.e.,
$$E\{\|f(\mathbf{y}_{\setminus i}) - y_i\|_2^2\}$$
. Such as Noise2Void [4], Self2Self [5]

[1] Lehtinen, J. et al. (2018). *Noise2Noise: Learning image restoration without clean data*. NeurIPS.

[2] Pang, T. et al. (2021). *Recorrputed-to-Recorrputed: Unsupervised Deep Learning for Image Denoising*. CVPR.

[3] Mansour & Heckel (2023). *Zero-shot Noise2Noise: Efficient image denoising without any data*. CVPR.

[4] Krull, A. et al. (2019). *Noise2Void: Learning denoising from single noisy images*. CVPR.

[5] Quan, Y. et al. (2020). *Self2Self with dropout: Learning self-supervised denoising from single image*. CVPR.

Equivalent with Supervised Loss.

- Noise2Noise:

$$\begin{aligned} E\{\|f(\mathbf{y}_1) - \mathbf{y}_2\|_2^2\} &= E\{\|f(\mathbf{y}_1) - \mathbf{x} - \mathbf{n}_2\|_2^2\} \\ &= E\{\|f(\mathbf{y}_1) - \mathbf{x}\|_2^2 + E\{\|\mathbf{n}_2\|_2^2\} - 2E\{(\mathbf{f}(\mathbf{y}_1) - \mathbf{x})^T \mathbf{n}_2\} \end{aligned}$$

- R2R, assume $\mathbf{n}, \mathbf{z}$ from the same distribution:

$$\begin{aligned} &E_{x,n,z}\{\|f(\mathbf{y} + \mathbf{z}) - (\mathbf{y} - \mathbf{z})\|_2^2\} \\ &= E_{x,n,z}\{\|f(\mathbf{y} + \mathbf{z}) - \mathbf{x}\|_2^2\} + \text{const, since } E_{(n,z)|x}\{f(\mathbf{x} + \mathbf{n} + \mathbf{z})(\mathbf{n} - \mathbf{z})\} = 0. \end{aligned}$$

- Powerful Blind Spot Scheme:

$$\begin{aligned} &E\{\|f(\mathbf{y}_{\setminus i}) - y_i\|_2^2\} \\ &= E\{(f(\mathbf{y}_{\setminus i}) - x_i)^2\} + E\{n_i^2\} - 2E\{(f(\mathbf{y}_{\setminus i}) - x_i)n_i\} \end{aligned}$$

Consider only AWGN, therefore the zero-mean and independent property will lead to the vanishing of the last term, which make the self-supervised loss equivalent to the supervised loss, theoretically with no biased prediction.

Self-supervised Denoising Methods

- Unsupervised denoising: using noisy dataset.
- Zero shot denoising at test time: using single noisy image.

Deep Image Prior (DIP)[1] is a pioneering work on zero-shot denoising, leveraging the inductive biases of convolutional neural networks (CNNs) to fit natural images more rapidly than random noise. The concept of the blind-spot principle can also be extended to zero-shot denoising at test time, e.g. Self2Self[2], ZS-N2N[3], MASH[4].

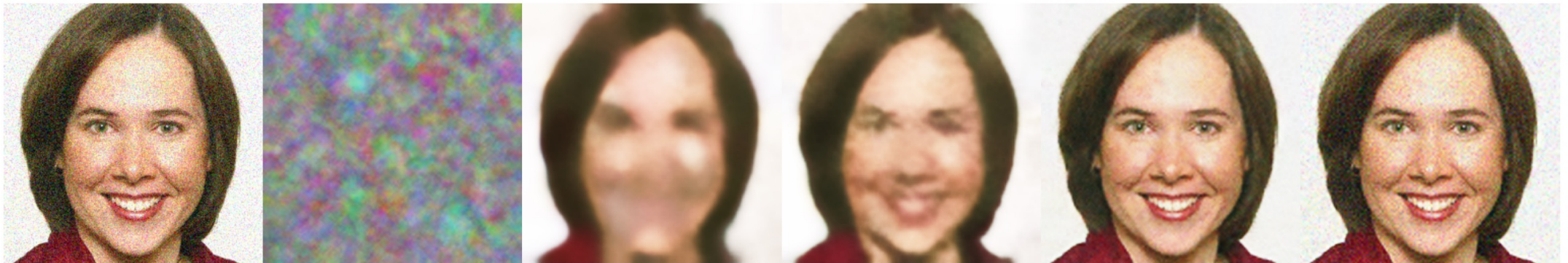
[1] Ulyanov, D. et al. (2018). *Deep Image Prior*. CVPR.

[2] Quan, Y. et al. (2020). *Self2Self with dropout: Learning self-supervised denoising from single image*. CVPR.

[3] Mansour & Heckel (2023). *Zero-shot Noise2Noise: Efficient image denoising without any data*. CVPR.

[4] Chihaoui, H. et al. (2024). *Masked and Shuffled Blind Spot Denoising for Real-World Images* CVPR.

Illustration of DIP (with early stopping)



Iter = 0

Iter = 25

Iter = 50

Iter = 500

Iter = 1500



Figures from Chihaoui, H., & Favaro, P. (2025). *Diffusion Image Prior*. arXiv.
<https://doi.org/10.48550/arXiv.2503.21410>

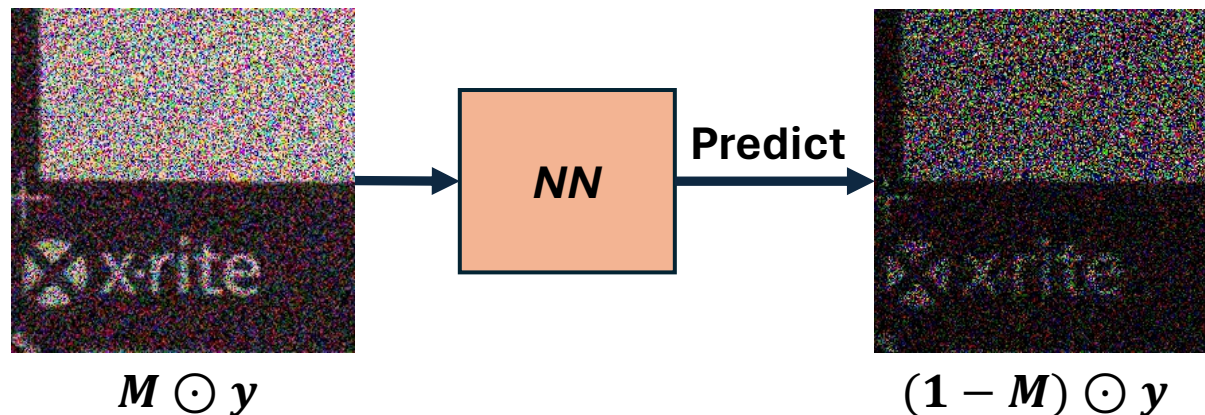
Different Blind Spot Schemes

Mask-based: directly mask some of the input and predict them by the rest pixels.

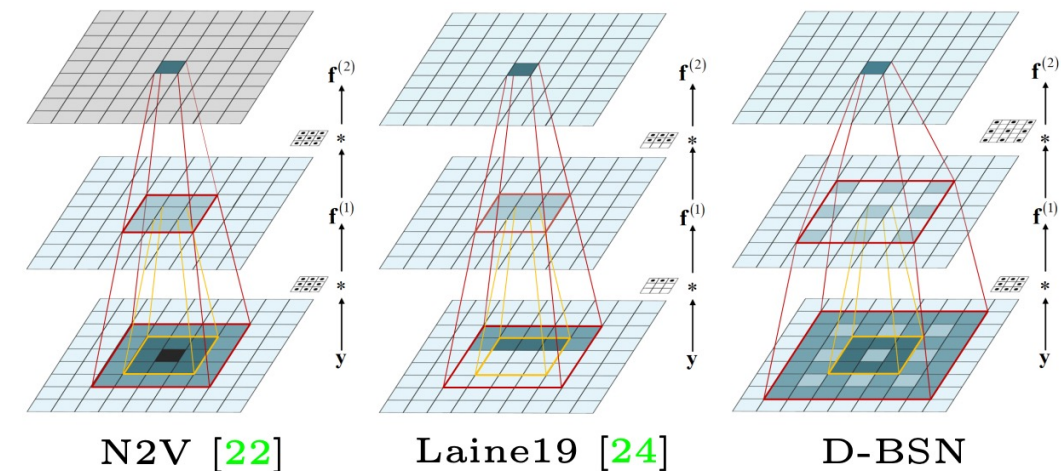
$$L = E|| (1 - M) \odot (f_{NN}(M \odot y) - y)||,$$

where M follows Bernoulli distribution.

Most mask-based methods are single image denoising.



Architecture-based: use designed NNs, e.g. D-BSN[1] uses central-masked kernel with following dilated convolution to avoid leak of the hidden pixel.



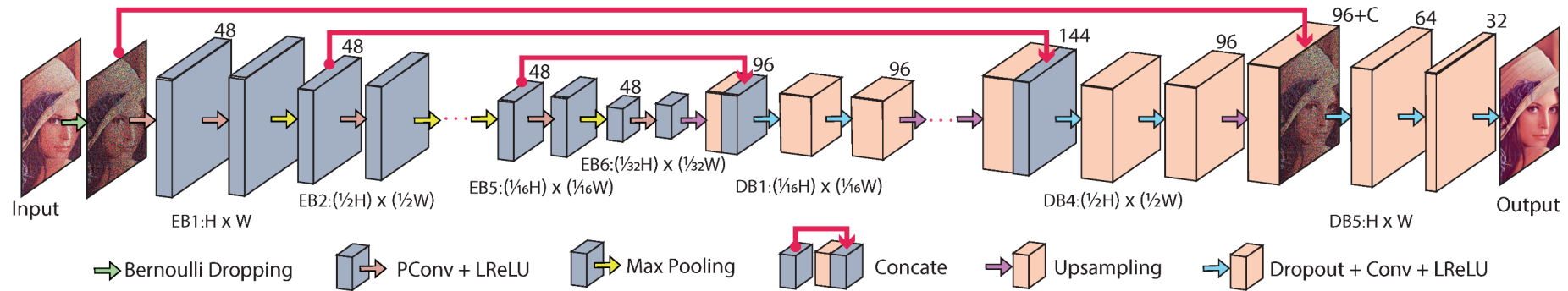
(a) Mechanisms of typical BSNs

Synthetic Noise Denoising: Self2Self[1]

- Decomposition of MSE:

$$\begin{aligned} E||f(\mathbf{y}) - \mathbf{x}||_2^2 &= E||E[f(\mathbf{y})] - \mathbf{x}||_2^2 + E||f(\mathbf{y}) - E[f(\mathbf{y})]||_2^2 \\ &= \text{Bias}^2 + \text{Variance} \end{aligned}$$

- Dropout-based Unet:



- Loss function: $L = ||(1 - M_i) \odot (f_i(M_i \odot y) - y)||_2^2$
- Inference with dropout by approximate MMSE estimator to suppress the variance:

$$\hat{x} = \frac{1}{T} \sum_{i=1}^T f_i(M_i \odot y)$$

[1] Quan, Y. et al. (2020). *Self2Self with dropout: Learning self-supervised denoising from single image*. CVPR.

Results

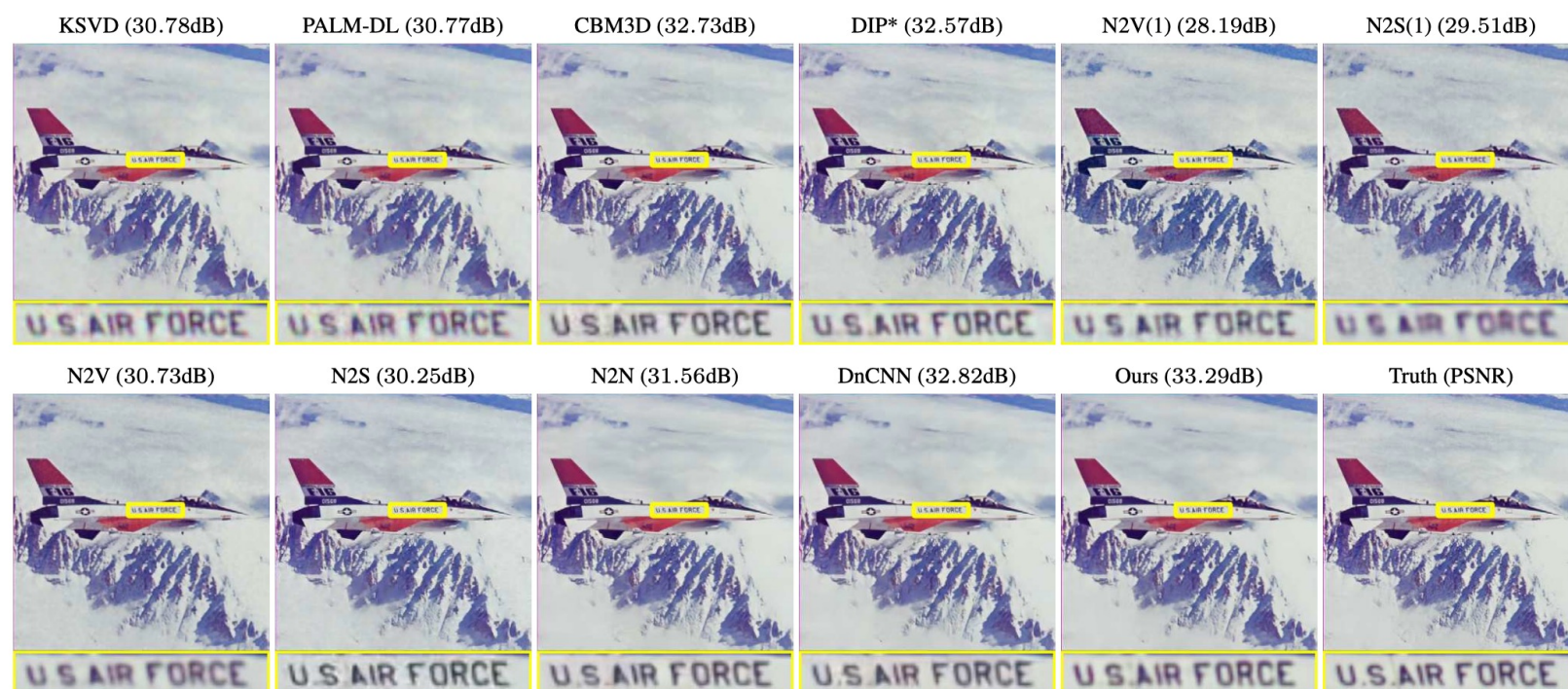


Figure 2. Visual results of blind AWGN denoising on image 'F16' of Set9 with noise level $\sigma = 25$.

Table 1. Average PSNR(dB)/SSIM($1.00E-1$) of AWGN removal results on Set9 and BSD68. The best results in all approaches are marked in bold, and the best ones in single-image-based approaches or dataset-based deep approaches are underlined.

Dataset	σ	Single-image learning or non-learning methods							Dataset-based deep learning methods			
		KSVD	PALM-DL	(C)BM3D	N2V(1)	N2S(1)	DIP*	Ours	N2V	N2S	N2N	(C)DnCNN
Set9	25	30.00/9.35	29.84/9.32	31.67/9.55	28.12/9.12	29.30/9.40	30.77/9.42	31.74/9.56	30.66/9.47	30.05/9.44	31.33/9.57	<u>31.42/9.56</u>
	50	26.50/8.70	26.64/8.70	28.95/9.22	26.01/8.75	27.25/9.04	28.23/9.10	29.25/9.28	27.81/9.12	27.51/9.05	<u>28.94/9.29</u>	28.84/9.25
	75	24.29/8.10	24.55/8.12	27.36/8.95	24.18/8.27	25.85/8.61	26.64/8.83	27.61/9.01	25.99/8.75	26.49/8.82	<u>27.42/9.05</u>	27.36/9.01
	100	23.12/7.70	23.18/7.67	26.04/8.68	23.55/7.80	24.67/8.48	25.41/8.58	<u>26.27/8.77</u>	25.37/8.58	25.46/8.57	26.45/8.86	26.30/8.78
BSD68	25	28.42/7.96	28.24/7.90	28.56/8.01	25.34/6.81	27.19/7.69	27.96/7.74	<u>28.70/8.03</u>	27.72/7.94	28.12/7.92	28.86/8.23	29.14/8.22
	50	25.08/6.53	25.09/6.49	25.62/6.87	23.85/6.18	24.53/6.42	25.04/6.45	<u>25.92/6.99</u>	25.12/6.84	25.62/6.78	25.77/7.00	26.20/7.15

ZS-N2N[1]

- Creating pairs by diagonally average downsampling.

- Residual Learning(learn the noise) in Noise2Noise:

$$D_{f_\theta}(y) = y - f_\theta(y)$$

$$L(\theta) = ||D_{f_\theta}(y_1) - y_2||_2^2$$

- Symmetric loss:

$$L_{res.}(\theta) = \frac{1}{2} (||D_{f_\theta}(y_1) - y_2||_2^2 + ||D_{f_\theta}(y_2) - y_1||_2^2)$$

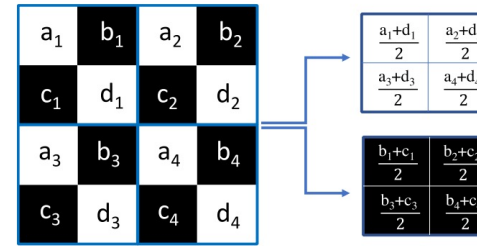
- Consistency loss (claim to prevent overfitting/early stopping):

$$L_{cons.}(\theta) = \frac{1}{2} (||D_{f_\theta}(y_1) - D_{f_\theta}(y)_1||_2^2 + ||D_{f_\theta}(y_2) - D_{f_\theta}(y)_2||_2^2)$$

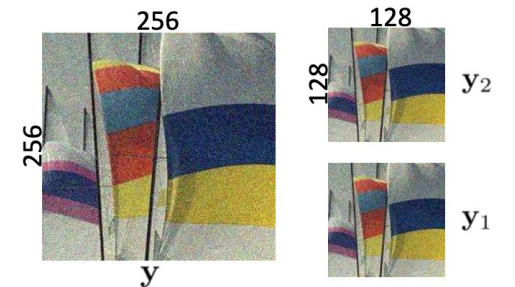
Total loss:

$$L_{total}(\theta) = L_{res.}(\theta) + L_{cons.}(\theta)$$

- Assuming that nearby pixels of clean image are highly correlated, while noise is independent and unstructured
- Downsample a noisy image into a pair of smaller images, which is an approximation of two noisy observations of the same clean image



$$\mathbf{k}_1 = \begin{bmatrix} 0 & 0.5 \\ 0.5 & 0 \end{bmatrix} \quad \mathbf{k}_2 = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



2D depthwise convolution, with stride 2

$$y_1 = y \circledast \mathbf{k}_1 \quad y_2 = y \circledast \mathbf{k}_2$$

Challenge of Real-world Denoising

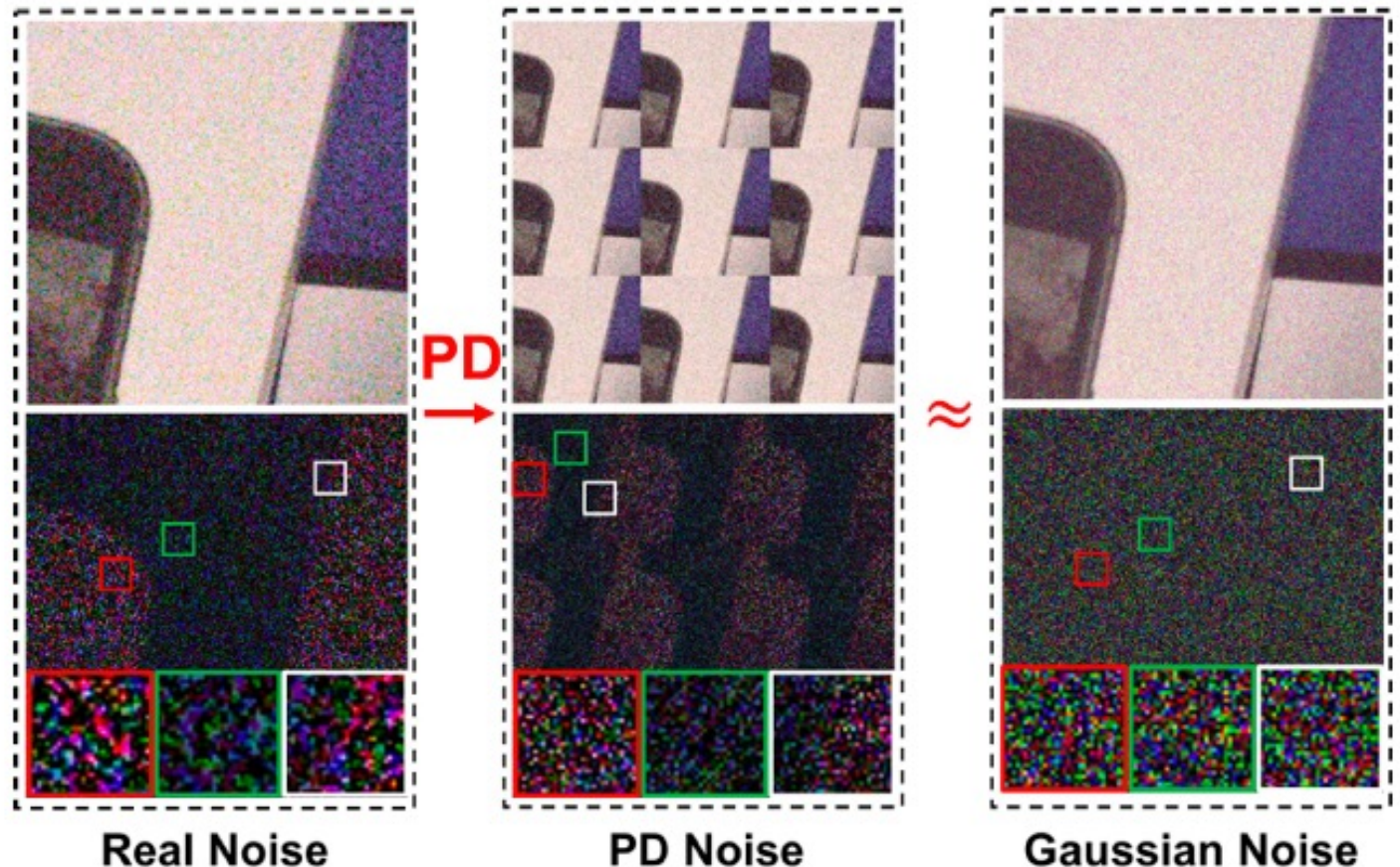
- **Multiple noisy captures** of the same scene are often impractical.
 - **Simulating real-world noise** accurately is difficult.
 - **Real-world noise** tends to be **locally correlated**.
-
- There is room for us to handle locally correlated real-world noise with only noisy datasets. Without facing first two challenges, blind-spot scheme utilizes the neighboring pixels to predict the invisible pixels, which has a great potential to extend to real-world denoising task.

Challenge of Blind Spot Scheme for Real-world Denoising

- When noise is spatially correlated, the neighboring pixels may carry similar noise patterns, leading to biased predictions.
- True value of pixels used for prediction should correlate closely to enhance the capability of network.

One Possible Way to Handle the Correlated Noise

- PD-Denoising[1] finds real-world noise in noisy image after Pixel-shuffling
Downsampling(PD)
becomes less correlated and similar to AWGN.
- But such PD operator could also lower the true pixel correlation.



Isolated Artifacts Caused by PD

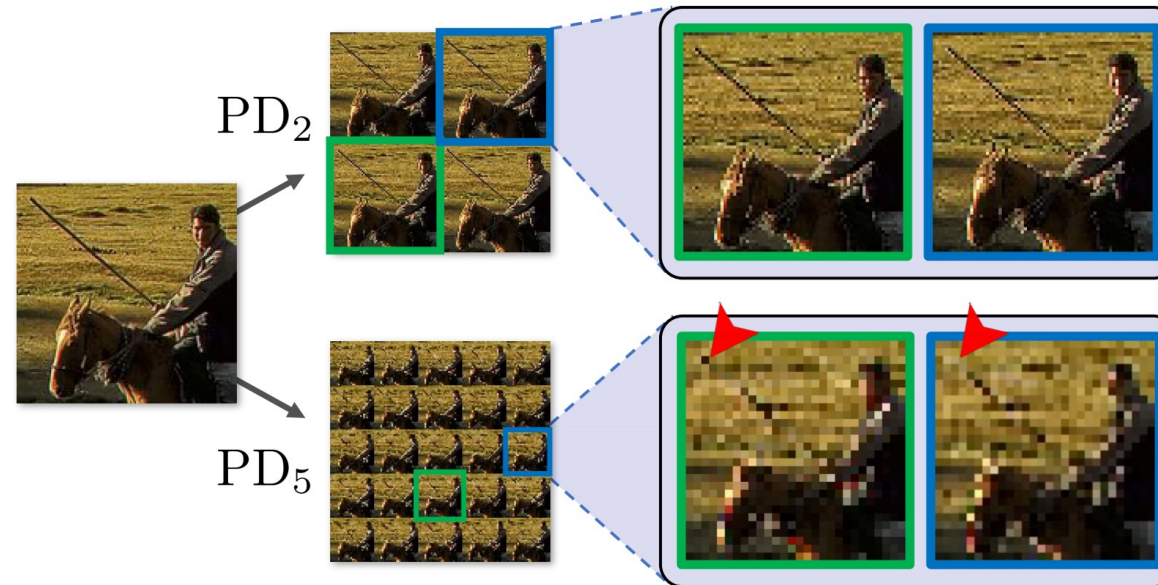


Figure 4. **Comparison between PD_2 and PD_5 .** Each operation decomposes the given image into 4 and 25 sub-images, respectively. In sub-images from PD_5 , we mark the aliasing artifact, *i.e.* a black dot, with **red**, which can be interpreted as noise for BSN. We note that the artifact does not appear in the **blue** sub-image.

Blind Spot Scheme for Real-world Denoise Based on PD

- Combined with blind spot scheme, if the noise correlation is decreased, then the self-supervised loss function could be less biased.
- Although PD could decorrelate the noise, it will also cause loss of structure correlation and lead to isolated artifacts.
- Numerous works build upon PD and aim to preserve structural correlations.

Asymmetric PD for Blind Spot (APBSN[1])

- Small stride 2 is not sufficient enough to decorrelate the noise, while large stride 5 will decrease the structure correlation (leading to isolated artifacts which could be removed by BSN)
- APBSN[1] propose a strategy to training using PD with stride 5 while testing using stride 2.

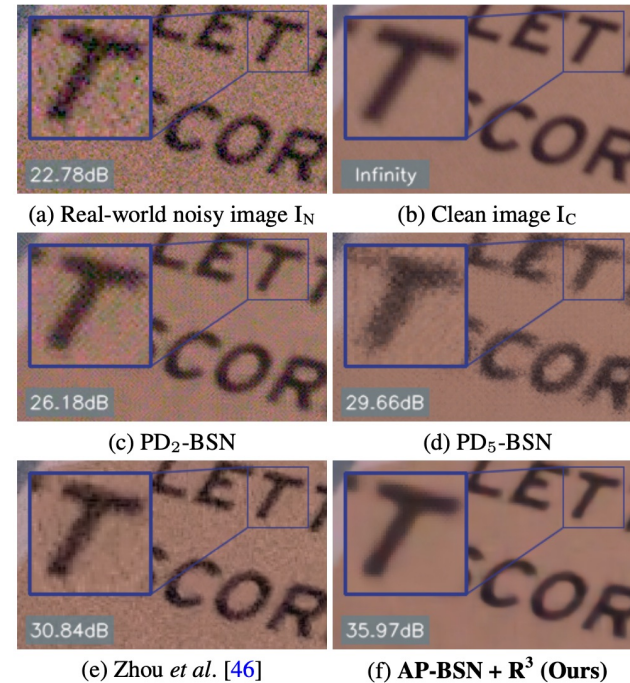


Figure 3. **Issues on PD_s -BSN when handling real-world noise.** (c) With a small stride factor, PD-BSN cannot remove noise from the input I_N . (d) With a large stride factor, PD-BSN destructs edge structures. (e) When AWGN denoiser meets PD [46], the model cannot completely remove real-world noise. (f) Our self-supervised approach delivers an accurate denoising result by overcoming the limitation of combining PD and BSN.

D-BSN Architecture Used in APBSN

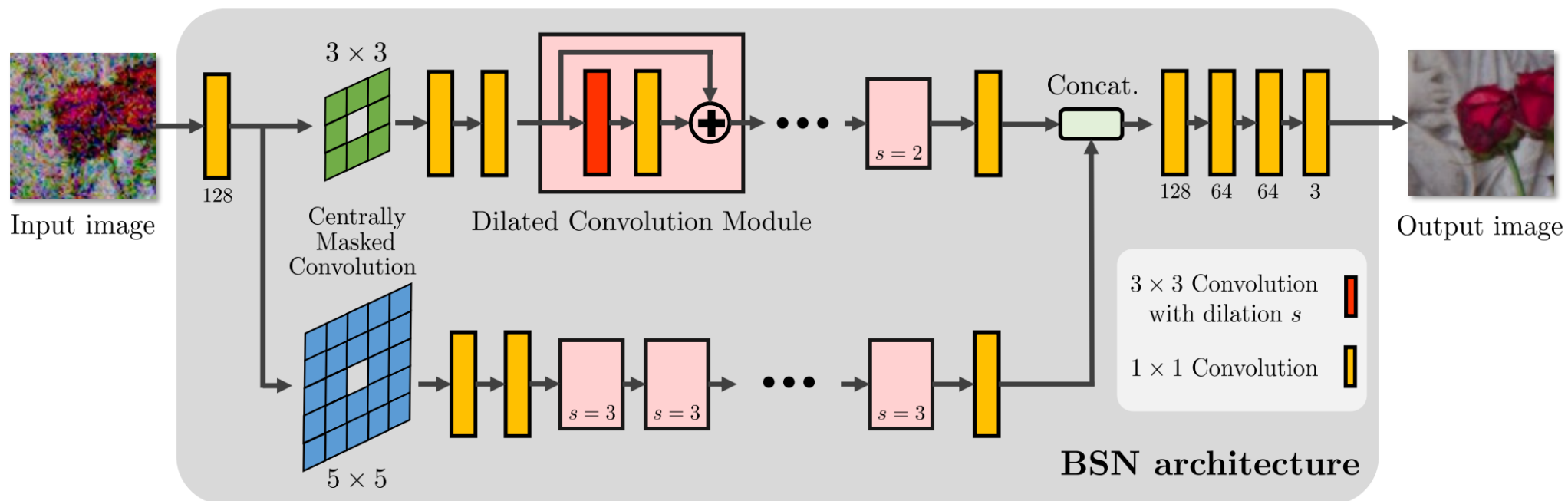


Figure S1. **Visualization of our BSN architecture.** We adopt 3×3 and 5×5 Centrally Masked Convolutions [38] to implement the blind-spot network. Each Dilated Convolution module (DC) contains one 3×3 dilated convolution with a stride s , where $s = 2$ and $s = 3$ are used for the upper and lower path of the network, respectively. For each path, we stack 9 DC modules. The number of output channels is denoted below each convolutional layer, where 128 is used by default.

Patch-unshuffle Downsampling in UNet

- PUCA[1] proposes the patch-unshuffle downsampling to avoid the leak of hidden pixels in UNet architecture, while simple downsampling fails.
- Such downsampling effectively utilizes multi-scale representation and expands receptive fields. (the advantage of UNet)

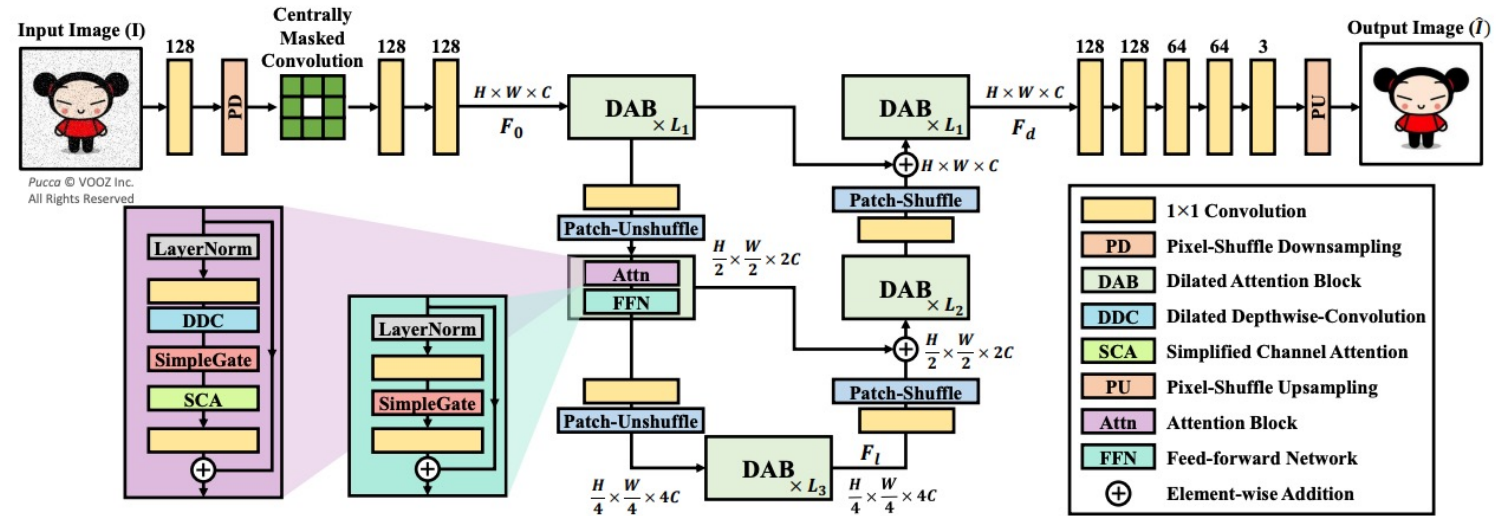


Figure 3: **Overview of PUCA.** Our method utilizes an encoder-decoder-based U-Net architecture. During encoding, we extract global context through patch-unshuffling, and during decoding, we combine local and global context through skip connections.

[1]Jang et al. (2023). *PUCA: Patch-unshuffle and channel attention for enhanced self-supervised image denoising*. NeurIPS. <https://arxiv.org/abs/2310.10088>

How Patch-unshuffle Prevents Leaking

- Patch size should be the multiple of stride of dilation to avoid the leak.

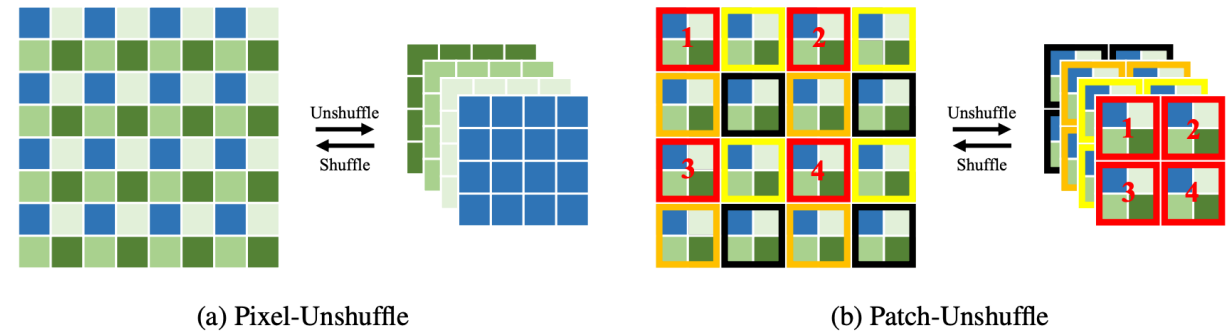


Figure 4: **Difference between pixel-unshuffle/shuffle and patch-unshuffle/shuffle.** The pixels except for blue are dependent on the central pixel of input, and only blue pixels are independent. We assume 2-dilated convolution is applied.

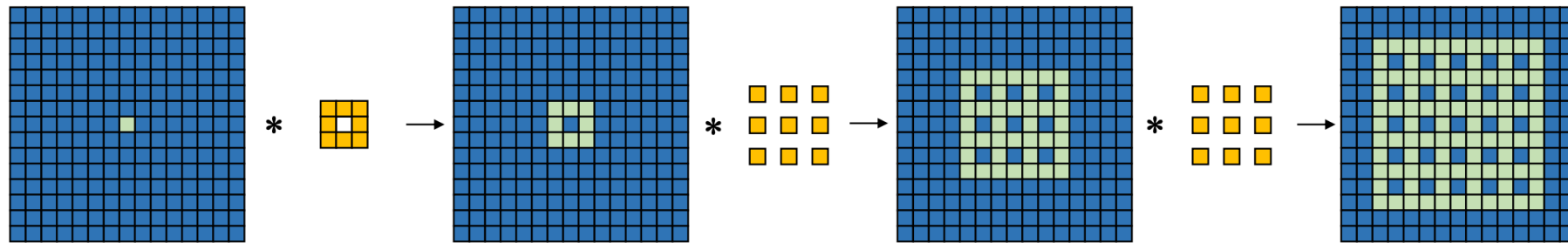


Figure 2: **Dependency between the input and output pixels with a centrally masked convolution and d-dilated convolutions.** The green pixels indicate the pixels that are dependent on the center pixel in input, while the blue pixels represent the area independent of the central pixel. The yellow pixels represent the convolution weights.

Blind Spot Scheme for Real-world Denoise Based on Local Pixel Shuffling

- From the conclusion of denoise method based on data augmentation e.g. R2R, underlying flat region in noisy image might share similar truth while different noise, which by local pixel shuffling could provide the paired noisy images.
- MASH[1] utilizes such intuition, use local variance and random local pixel shuffling to denoise.

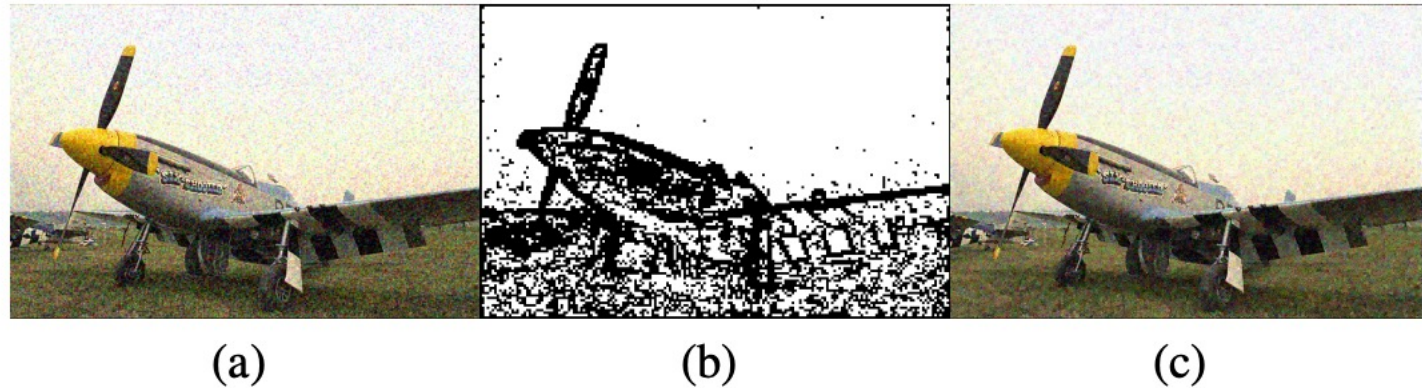


Figure 4. (a) Original noisy image (b) Mask capturing region flatness derived from pseudo-clean prediction (c) Noisy image with local pixel permutation on flat regions.

Recent Results on Real-world (SIDDD)

- There are still room for zero-shot real-world denoising.

	Methods	SIDD-Valid	SIDD-Bench	F
Non-local	BM3D '07	25.65	25.65	
Self-supervised	Noise2void '19	27.06	26.99	
	CVF-SID '21	34.81	34.71	
	R2R '22	35.04	34.78	
	APBSN '22		36.91	
	Selfformer '24		37.67	
Zero-shot	DIP '18	32.11		
	Self2Self '21	29.46	29.51	
	ScoreDVI '23	34.75	34.60	
	MASH '24	35.06	34.78	
	INS, '25	35.31	35.05	



Thank you!

